Hongyue Yu

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EDUCATION

Beihang University (BUAA) Sep 2024 – Jul 2027 (*expected*)
Master of Engineering National College for Excellent Engineers

- Advisor: *Prof. Yuan Yuan*

Beihang University (BUAA) Sep 2020 - Jul 2024
Bachelor of Computer Science and Engineering School of Computer Science and Engineering

- GPA: 3.74/4.0
- First Class Scholarship of Beihang University(2021)
- Competition Scholarship of Beihang University(2022)

RESEARCH INTERESTS

Automatic software engineering; AI for software engineering.

SELECTED PUBLICATIONS(Full Publications in Personal Homepage)

1. **Hongyue Yu,** Kefan. Li, et al.“TDD-Agent: Test-Driven Reasoning for Code Generation,” *Conference on Empirical Methods in Natural Language Processing* (**EMNLP 2026**), May 2026. [arxiv]

✓ Proposed **TDD-Agent**, a test-driven code generation framework that **treats executable tests as reasoning artifacts**, requiring LLMs to **specify expected behavior before implementation** and then **co-refine tests and code** through execution feedback. Unlike prior self-testing approaches that treat generated tests as fixed validators, TDD-Agent allows the agent to correct both implementation defects and flawed test assumptions, continuously aligning specification with implementation. Evaluations on **LiveCodeBench** and **RepoEval** show consistent improvements over reasoning-, retrieval-, and agent-based baselines across multiple LLMs.

2. Kefan. Li**, Hongyue Yu**, and Yuan Yuan. “ Escaping the Self-Repair Trap: Improving Test Oracle Generation via Dual-Context Awareness,” in *IEEE/ACM International Conference on Automated Software Enginerring* (**ASE 2026**), Mar 2026. [arxiv]

✓ Identified the **Self-Repair Trap**, a failure mode in LLM-based test oracle generation where iterative execution-guided repair increasingly favors **easier-to-satisfy but less fault-revealing** assertions. Proposed **DCAware**, a computationally efficient **non-iterative dual-context** framework combining **Denoised Static Program Context** with **Intent-Driven Dynamic State Extraction** for targeted runtime evidence. DCAware improves both execution success and fault-revealing capability while requiring one-quarter of the API cost and up to 77% less execution time than iterative agentic approaches.

3. Kefan Li, Yuan Yuan, **Hongyue Yu**, et al. “ CoCoEvo: Co-Evolution of Programs and Test Cases to Enhance Code Generation,” *IEEE Transactions on Evolutionary Computation*(**TEVC**), May 2025. [arxiv]

✓ Proposed **CoCoEvo**, an **LLM-based co-evolutionary framework** that jointly evolves programs and self-generated test cases without predefined test suites. CoCoEvo combines **LLM-based crossover and mutation**, dynamic crossover-rate scheduling, **coverage-guided test evolution**, and **Pareto-based multi-objective selection** over test confidence and discrimination to preserve both reliable and fault-revealing tests. Co-evolving code with its testing supervision consistently achieves the best Pass@1 across fixed-test repair and agreement-based baselines on the LeetCode-Contest benchmark.

4. Kefan Li**, Hongyue Yu**, Yuan Yuan, et al. “Beyond Fixed Tests: Coevolution of Code and Behavioral Constraints”, submitted to *IEEE Transactions on Software Engineering*(**TSE**). [paper]

✓ Proposed **SWE-CoEvo**, a repository-level coevolutionary framework that reframes issue resolution as a **coupled search over candidate repairs and executable behavioral constraints**, jointly evolving

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source-code patches and issue-revealing test patches. SWE-CoEvo integrates **AST-aware source/test context construction**, structured repository edits, **trace-guided semantic operators**, and execution-based cross-evaluation through a **code–test agreement matrix and dual-agreement fitness**, enabling repair and test populations to mutually guide refinement. It achieves the best overall performance on **SWE-bench** and **SWT-bench** while producing both artifacts in a single run.

5. Kefan Li**, Hongyue Yu**, et al., “Passing is Not Roof: How LLM Agents Decide from Self-Generated Tests”, submitted to *International Conference on Software Engineering* (**ICSE 2027**). [paper]

✓ Conducted a **trajectory-level study of how LLM coding agents reason from self-generated test feedback**, independently labeling intermediate code and test artifacts to distinguish **failure attribution** after failed tests from **decision calibration** after passed tests. Revealed a strong asymmetry: agents often continue repairing after failures but frequently **over-trust passing self-generated tests**, proceeding with unresolved code, non-reproducing tests, or mutually incorrect code–test pairs. Introduced **final wrong agreement** to quantify this false confidence and evaluated an **independent issue-alignment guard** that reduces wrong-proceed decisions and wrong agreements without relying on ground-truth patches or tests

INTERNSHIPS

Tokfinity AI, Beijing, China Nov 2025-Mar 2026

- Proposed InfCode, a multi-agent framework for repository-level issue resolution in C++ and Java, introducing a dual-layer retrieval architecture that combines **intent-guided semantic retrieval** with **structural code search** for coarse-to-fine fault localization; designed **Intent and Entity Graphs** to bridge natural-language descriptions with implementation-level semantics and code dependencies, enabling precise code navigation beyond lexical search and achieving 1st place on both the Java and C++ leaderboards of Multi-SWE-bench.

4Paradigm, Beijing, China Apr 2025-Jul 2025

- Trained an attention-based classification head on CLIP for video moderation, achieving 4-way concurrency with sub-4s latency in a resource-constrained environment, with a 12.7% false-negative rate and 17% false-positive rate.

SKILLS

Languages: English (proficient), Mandarin Chinese (native) **Skills:** Python, Java, C/C++, C#, etc
Tech Stack: Pytorch, Vue, MySQL, Git, Docker **Enviroment:** Linux, macOS, Windows